GlassQuill

Explainable AI-Text Detection

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Course
Applications and Use Cases
of Machine Learning (TE7483)
ESE case study

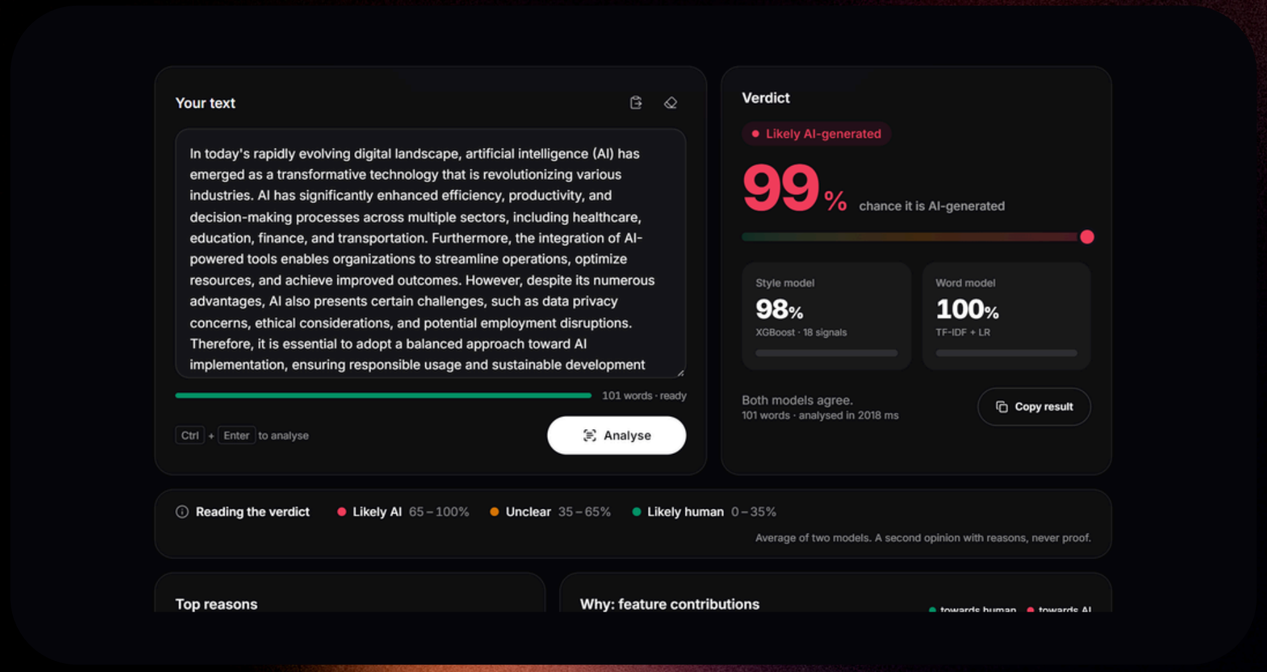
Code and app
github.com/ThatOneRahul/glassquill
AIML Department, SIT Pune

Agenda

About 12 minutes in seven parts, then a live demo of the app and questions

01 Problem	02 Data and cleaning	03 Features	04 Models
05 Implementation	06 Results and robustness	07 Explanations and conclusion	

See the evidence



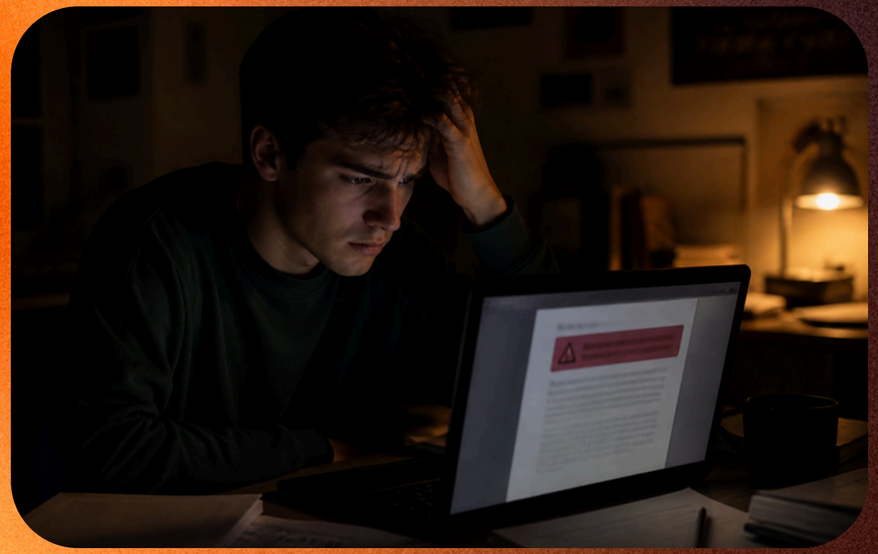
97.4%

accuracy on 4,829 held-out texts with only 2.0% of human texts flagged by mistake. Every verdict comes with the writing signals and the exact words behind it.

Problem Statement

Why it matters

A wrong flag accuses a real person



>50%

Liang et al. 2023

of non-native essays flagged as AI

Colleges and newsrooms use AI-text detectors that give a bare score. A student who is wrongly flagged cannot question it. GlassQuill gives a verdict with reasons and keeps false flags as low as possible.

Research Questions

01

RQ1 Signals

Which simple signals like perplexity, sentence rhythm, vocabulary and punctuation separate human and AI text? How strongly?

02

RQ2 Glass box

How close can explainable models (XGBoost with SHAP) get to a fine-tuned DistilBERT black box?

03

RQ3 + RQ4 Trust

RQ3 Does it hold up on unseen LLMs and on paraphrased AI text?
RQ4 Do two explanation methods, SHAP and LIME, agree?

Objective: explain every verdict and keep humans wrongly flagged as AI low

4 groups

Teachers want a second opinion with reasons. Students want a fair way to question a flag. Platforms need screening with an audit trail. Researchers need a clear baseline.

50-400

words of English text and one label: human (0) or AI (1). Domain: education and content moderation.

- Many genres, 17 domains
- Not other languages or code
- Not mixed human + AI text

Use Cases

Who needs it

Datasets

Texts after cleaning	Domains, 50/50 in each	Unseen generators (MAGE)
31,984	17	27

HC3 (ChatGPT vs human answers to the same questions), HAP-E (GPT-4o and Llama-3 in 6 genres) and M4GT (GPT-4, Mixtral, Gemma, Cohere and more). Split 70/15/15 by question or document: 22,371 train, 4,784 validation and 4,829 test texts. MAGE (2,000 texts) is kept only for the out-of-domain test.

Removing Shortcuts

99.3%

- Tokenizer leftovers
- Markdown and quotes
- Unbalanced domains

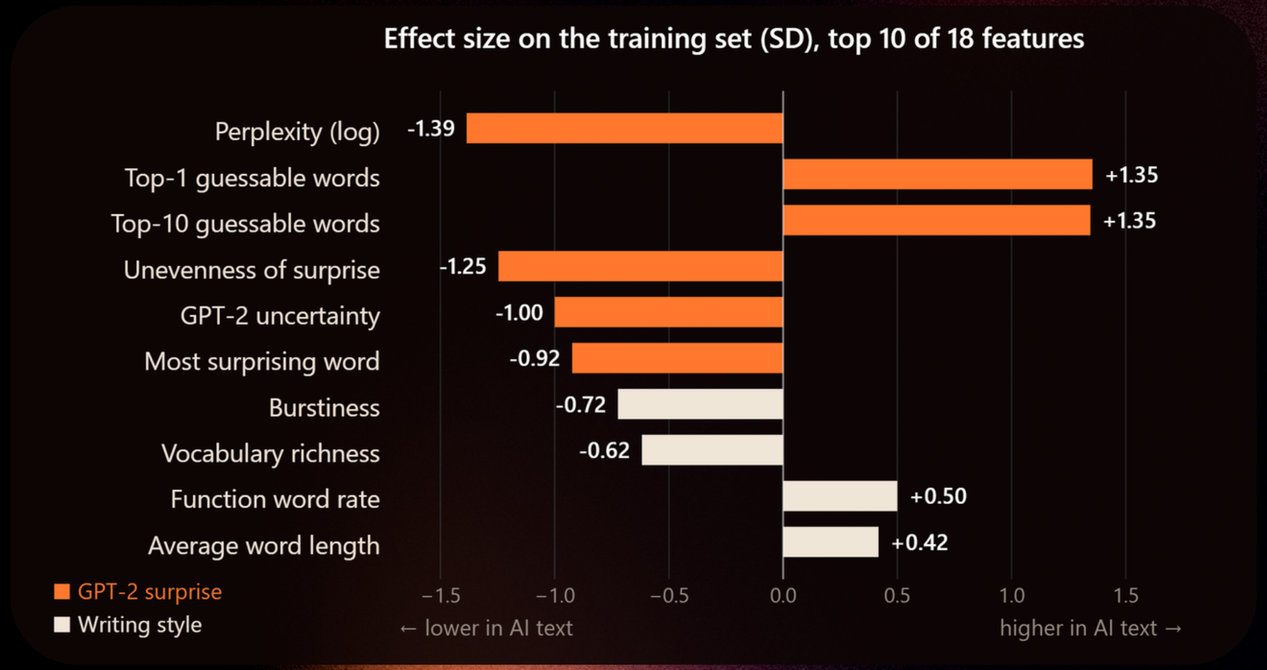
The first full run scored 99.3%. That was too good, so we looked for shortcuts: the models were spotting formatting, not writing. After cleaning every source the same way the score fell to a believable level.

What we fixed

Tokenizer splits like "do n't" were in 34% of HC3 human answers but only 0.02% of AI answers. URL_0 tags and spaced brackets too.

HC3 open_qa had 20 human vs 764 AI texts, so the topic gave away the label. Now every domain is 50/50 and splits are by question.

Writing Signals



18 features

6 come from GPT-2 and measure how easy each word is to predict. AI text is easier to predict, so perplexity is the strongest signal. The other 12 cover writing style: sentence rhythm, vocabulary, punctuation and readability.

Pipeline

run_pipeline.py runs the whole study in 8 cached stages with seed 42. Any stage can be rerun on its own and a quick mode reproduces everything on a sample in about 10 minutes.

01 Load 3 corpora	02 Clean shortcuts	03 Balance and split	04 Extract 18 features
05 Train and tune 4 models	06 Pick the verdict rule on validation		07 Test once, then explain

Four Models

01

Glass-box baselines

A perplexity threshold on GPT-2 alone, cut-off picked on validation. TF-IDF with logistic regression on 1-2 word features, tuned by 5-fold GroupKFold.

02

XGBoost on style

400 trees on the 18 features. Monotone limits where theory and data agree, so it cannot learn odd rules. Explained exactly with TreeSHAP.

03

DistilBERT black box

Fine-tuned for 2 epochs on raw text. The strongest single model but a black box, so we keep it only to compare against.

Verdict = mean of XGBoost and TF-IDF + LR, picked on validation. Unclear band: 35 to 65%

Built and Tested

28 tests

Python 3.11 with PyTorch, scikit-learn, XGBoost, SHAP and LIME. FastAPI serves a React site. The trained models are in the repo, so the app runs without training. A full study run takes about 1.5 hours on an RTX 3050 laptop GPU.

The app never stores the text. It checks the input is English prose, rate limits each IP and sends strict security headers.

Run it

```
python server.py
```

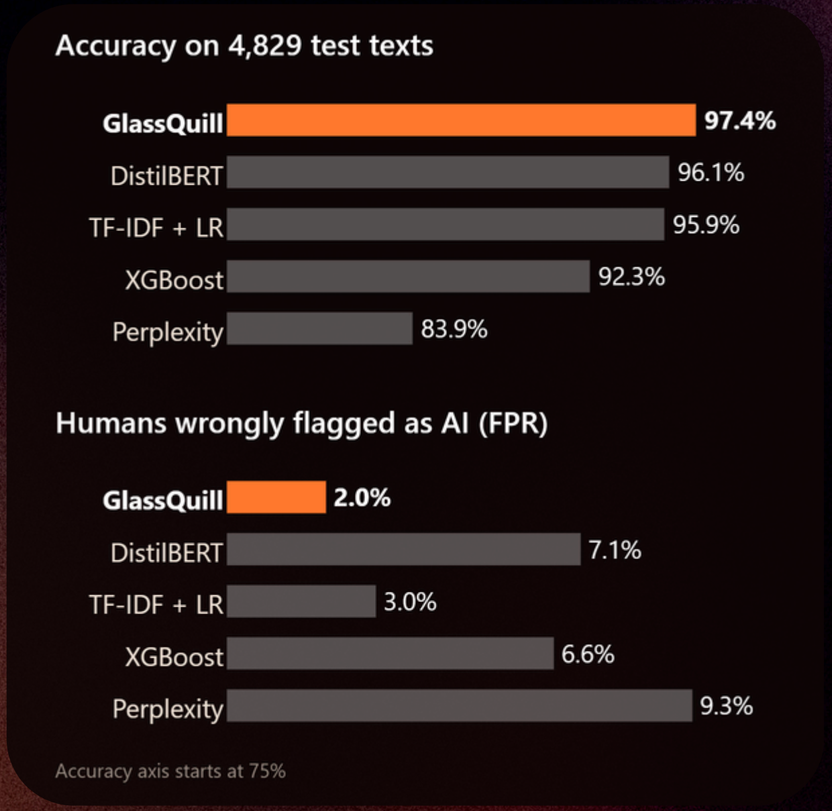
```
glassquill/
├── run_pipeline.py
│   8 cached stages, seed 42
├── server.py
│   FastAPI, input checks, rate limit
├── src/data.py
│   load, clean, balance, split
├── src/features.py
│   18 features (GPT-2 + style)
├── src/models.py
│   threshold, TF-IDF + LR, XGBoost
├── src/train_distilbert.py
│   black-box comparison
├── src/explain.py
│   SHAP, LIME, word weights
├── src/detector.py
│   one text in, verdict out
├── web/
│   React 19 + TypeScript site
├── tests/
│   28 API tests
└── models/ reports/
    trained models, tables, figures
```

Test Results

97.4%

accuracy for the GlassQuill verdict, the mean of XGBoost and TF-IDF + LR. F1 0.974 and ROC-AUC 0.995.

4,829 test texts



2.0%

of human texts wrongly flagged as AI. DistilBERT flags 7.1%, over three times more.

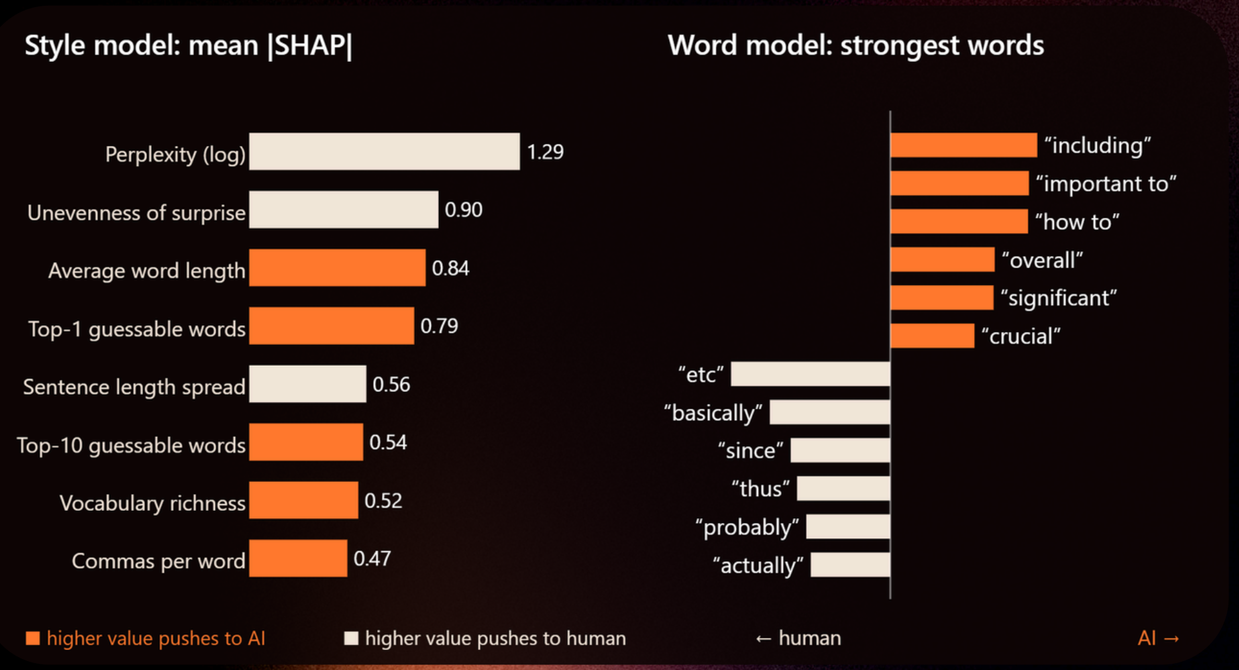
- HC3: F1 0.987
- HAP-E: F1 0.981
- M4GT: F1 0.956

Robustness

Accuracy on unseen MAGE	GPT-4 texts still caught	Caught after paraphrasing
66.4%	91%	93%

Every model drops to 61-66% on 27 generators it never saw, but GlassQuill keeps the lowest false positive rate (6.8% vs 17.6% for DistilBERT). Chat models are still caught. Small open models like OPT, BLOOM and GPT-J are not. After T5 rewrites every sentence GlassQuill still catches 93% of AI texts (98% before).

Showing



0.06

is how much SHAP and LIME agree on DistilBERT's top words. LIME also found only 51% of the true top words of the linear model. So the app explains with exact SHAP and exact word weights instead.

Reasons

SWOT Analysis

01

Strengths

97.4% accuracy with only 2.0% false flags. Every verdict has reasons a teacher can check. Tested, reproducible and private by design.

02

Weaknesses

66.4% on unseen generators. Small open models like OPT, BLOOM and GPT-J go undetected. English only and no mixed human + AI texts.

03

Threats

Paraphrasing tools and heavy editing. Plain writing by non-native writers can look AI-like. New LLMs can make results go stale.

Opportunities: Binoculars features, sentence-level detection and a threshold set for 1% FPR

Conclusion

4 / 4

- RQ1 Perplexity leads
- RQ2 Glass box wins
- RQ3 Partly robust

Two explainable models averaged beat a fine-tuned DistilBERT with under a third of its false flags. SHAP and LIME barely agree (RQ4), so we explain with exact methods. Removing dataset shortcuts mattered most.

Next steps

A threshold for a fixed 1% FPR, calibrated scores and sentence-level detection for mixed texts.

Test on RAID and newer generators and check fairness on non-native essays. A detector score is a second opinion, never proof.

Thanks Questions? & Everyone

- github.com/ThatOneRahul/glassquill
- Live demo of the detector next